

Qiyuan(Vincent)Zhu

Tel: 2154303398 | Email: qiyuanzhu522@gmail.com | <https://www.linkedin.com/in/qiyuan-zhu-6171b9249/> | GitHub: <https://github.com/vincentvicente>

EDUCATION

Northeastern University

Master of Computer Science

Sep 2023-May 2026

University of Pennsylvania

Master of Public Administration

Sep 2022-May 2023

TECHNICAL SKILLS

Programming Languages: Java, JavaScript/TypeScript, Python

Frameworks/Tools: Backend - Springboot/Node.js, ORM – Hibernate/Mybatis/Prisma; Micro Services; API-SOAP/REST/RPC, Converter - Jackson2; Dependency – Maven/Gradle; Git – GitHub, Bitbucket; Docker; Linux - Command, CronJob; CI/CD - AWS CodeCommit, CodeBuild, ECR, CodeDeploy; Message Queue - Kafka, RabbitMQ, AWS Kinesis/SNS/SQS; Relational Database - MySQL, PostgreSQL, AWS RDS; NoSQL - MongoDB, AWS DynamoDB; Serverless - AWS Lambda; Cache - Redis; Unit Test - Junit; Monitoring & Dashboard - Splunk

Cloud Platform: AWS(S3, EC2, RDS, DynamoDB, Lambda, Kinesis, SNS, SQS, RedShift, Glue, IAM, EMR, CodeCommit, CodeBuild, ECR, CodeDeploy etc.), Azure, GCP

AI Development Tools: Claude Code, Cursor, ChatGPT, GitHub Copilot

PROFESSIONAL EXPERIENCES

Shawk Solution

Software Engineer Intern

Aug 2025-Oct 2025

- Developed a full-stack **RAG** document QA system by integrating **FastAPI**, **SQLite**, and **React/TypeScript**, supporting **1,000+** PDF/DOCX uploads and achieving **95%** accuracy in content extraction.
- Built 8+ **RESTful APIs** in FastAPI for document ingestion, parsing, text chunking, vector embeddings, and semantic search, delivering **<1.5s** average query latency.
- Created a responsive **React** frontend with **TypeScript**, featuring drag-and-drop upload, document list, and real-time chat, tested with **50+** active users.
- Implemented a robust **service pipeline** (document_service, embedding_service, rag_service) managing **50,000+** indexed text chunks, enabling scalable vector search and offline fallback testing.

Artisk Inc.

Software Engineer Intern

Feb 2025-Apr 2025

- Built a **Multi-Agent Orchestration System** using **Coze AI** and **Claude 3**, implementing a **Dynamic Prompt Pipeline** that increased design intent recognition accuracy by **45%** through structured metadata extraction and Few-shot learning.
- Architected a high-concurrency **Asynchronous Task Queue** with **Meteor** and **MongoDB**, utilizing **WebSocket**(DDP protocol) for real-time status pushing; eliminated UI blocking for long-running AI operations and managed complex concurrency control.
- Integrated a **Multi-model Generative Pipeline**, implementing a **closed-loop feedback system** that captures *modelRevisedPrompt* to maintain “Visual Memory” across multi-turn design iterations.
- Developed a professional-grade **Interactive Branding Editor** with **Fabric.js**, supporting **vector manipulation** and **AI-powered Inpainting**, allowing users to complete end-to-end brand identity creation within **60 seconds**.

PROJECT HIGHLIGHTS

Video RAG Agent System

Jan 2026-Feb 2026

- Architected an AI-powered **Video RAG agent** by building **3 MCP** servers with **12 custom tools** and **5 Claude AI skills**, enabling autonomous video processing, knowledge extraction, and natural-language Q&A over video content.
- Engineered an end-to-end data pipeline: video download, audio extraction (**FFmpeg**), speech-to-text transcription (**OpenAI Whisper**), semantic chunking (**NLP/TF-IDF**), and vector embedding (**OpenAI Embeddings**), transforming unstructured video into a searchable knowledge base.
- Implemented a **vector similarity search engine** with **cosine-similarity ranking** and **metadata filtering**, allowing users to semantically query across multi-video libraries and generate AI-driven topic summaries with source attribution.
- Reduced transcription costs by **95%** by integrating a local Whisper model as a hybrid alternative to the OpenAI API, enabling processing of **50+** hours of video content at **~\$0.02/hour** for embeddings only.

AI Autonomous Agent Platform

Sep 2025-Oct 2025

- Developed a full-stack AI agent platform using **Java Spring Boot** and **Vue.js**, integrating **Spring AI** and **LangChain4j** for dynamic tool-calling, contextual reasoning, and RAG pipelines across **10+** specialized tools, processing knowledge bases with thousands of documents.
- Implemented **RESTful APIs** for agent interaction, retrieval-augmented Q&A, and tool execution, managing a vector database and achieving **< 500 ms** average response time through optimized query pipelines and PGVector integration.
- Built an autonomous reasoning workflow based on **ReAct** and **MCP protocols**, enabling the agent to plan, search, and generate multi-step outputs with self-correcting decision loops and structured memory persistence.
- Designed a real-time conversational UI with **Vue.js** featuring streaming token output, multi-turn dialogue management, and interactive tool-calling visualization, enabling users to monitor agent reasoning steps and inspect retrieved sources in context.